“IOT BASED FLOOD DETECTION AND DAM CONTROLLER SYSTEM”

PROJECT REPORT

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE AWARD OF DIPLOMA IN ELECTRONICS AND TELECOMMUNICATION ENGINEERING

A CAPSTONE SUBMITTED BY

MAHEK TAJUDDIN SAYYAD.

SNEHA RAMESH KSHIRSAGAR.

PRATIKSHA SIDDHARTH VAKALE.

UNDER THE SUBJECT OF CAPSTONE (22060)

GUIDE MR.A.S.CHAVAN

DEPARTMENT OF ELECTRONICS AND TELECOMMUNICATION
GOVERNMENT POLYTECHNIC KARAD (2023-2024)

GOVERNMENT POLYTECHNIC, KARAD.

CERTIFICATE

This is to certify that,

Miss.Mahek Tajuddin Sayyad (2100100139)

Miss.Sneha Ramesh Kshirsagar (2000100127)

Miss.Pratiksha Siddharth Vakale (2000100157)

of final year Electronics & Telecommunication Engineering students have completed project of final year having title

"**IOT BASED FLOOD DETECTION AND DAM CONTROLLER SYSTEM**"

during academic session 2023-2024 as a part of project work described by Government Polytechnic, Karad for partial fulfilment for the Diploma in Electronics & Telecommunication Engineering in the sixth semester.

The project work is the record of student's own work under my guidance and to my satisfaction.

Mr.A.S.Chavan	**Mr.B.V.Karade**	**Dr. K.M.Bagwan**
(Guide)	(Head of Department)	(Principle)

GOVERNMENT POLYTECHNIC KARAD

CANDIDATE DECLERATION

I hereby certify that the work which is being presented in the project report entitled

“Iot Based Flood Detection & Dam Controller System”

By us in partial fulfilment of requirement for the award of diploma in Electronics and Telecommunication Engineering submitted in the department of Electronics and Telecommunication Engineering is an authentic record of our own work carried out during EVEN (2023-24) guided by Mr.A.S.Chavan.

Signature and Name of the Students

Mahek Tajuddin Sayyad

Sneha Ramesh Kshirsagar

Pratiksha Siddharth Vakale

This is to certify that the above statement made by the candidate is correct to the best of our knowledge.

Signature of the Guide

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We are also grateful to team of "Automatic traffic light controller" for their highly encouraging and co-operative attitude. We express our friends and parents for their moral support during project.

Place- Government Polytechnic, Karad

Date- / / 2024

Your sincerely: -

Mahek Tajuddin Sayyad
Sneha Ramesh Kshirsagar
Pratiksha Siddharth Vakale

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ABSTRACT

Flood is one of the natural disasters that cannot be avoided. It happens too fast and affected so many lives and properties. Before this, most of the existing system that has been developed are only focus on certain areas. Other than that, majority of the public cannot monitor and have no idea when the flood going to be happened since they do not have any information and data about the weather condition. By having Smart IoT Based Flood Detection & Dam Controller, this will solve all the drawbacks of the existing system. The proposed system is suitable for cities and village areas. Furthermore, if the public has an internet access, they can monitor what is happening and predict if there is any upcoming flood at the web server. The proposed system is a low cost in design and easy for maintenance. This project will update the water level at the web server and the system will issue an alert signal to the citizens for evacuation so that fast necessary actions can be taken.

Chapter-1 (Introduction)

1.1 Introduction

The world's weather is changing rapidly due to effect from mankind activities such as pollutions, trees cutting, vehicle gas emission etc. Floods are the most usual damaging natural disaster that cause significant harm to life, property, and economy. Scientists estimate by 2030, if sea level rises by 4-Inches, it could potentially cause the severe flooding in many regions of the globe. This project implies a flood warning structure that can detect the water level and measure the speed of the increase in water level. To give the nearby peoples an earlier notification to evacuate before the water rises to the dangerously high level, the measured result is sent as an alert on a mobile through Short Message Service (SMS). This prototype is designed on a IOT platform, where data from the sensor is stored at the mini-processor and alert is generated & sent as SMS to a smartphone. This prototype system is implemented in an experimental setting in two different environments to test its effectiveness. And also, we can monitor the temperature and humidity for climate record.

- The System Works In 3 Steps:
- Monitoring water level in Dam/River.
- Monitoring of rain & climate changes for info purpose.
- Display the measured data & camera live feed on thing speak and YouTube live streaming.
- If the level of water increases above defined value SMS & Vocal Alert via Speaker is provided.

1.2 Motivation

The motivation behind developing an IoT-based flood detection and dam controller system stems from the urgent need to enhance our ability to mitigate the impacts of floods effectively. Here are several key motivations driving this initiative:

1. Early Warning and Rapid Response: By deploying IoT sensors in flood-prone areas, we can gather real-time data on water levels, rainfall intensity, and weather conditions. This data enables us to provide early warnings to communities at risk, allowing them to take proactive measures such as evacuation and securing assets before the floodwaters rise. Rapid response is crucial in minimizing casualties and reducing property damage during flooding events.

2. Precision Monitoring and Prediction: Traditional flood monitoring systems often rely on manual measurements or stationary sensors with limited coverage. In contrast, an IoT-based system offers precision monitoring with a network of interconnected sensors distributed across various locations.

These sensors collect data continuously, enabling us to analyse trends, predict flood patterns, and make informed decisions regarding flood management strategies.

3. Optimal Dam Control and Water Release: Dams play a critical role in flood control by regulating water flow downstream. However, manual dam operations may not always align with changing weather conditions or the needs of downstream communities. By integrating IoT technology into dam control systems, we can automate the process of water release based on real-time data inputs such as upstream water levels, rainfall forecasts, and river flow rates. This dynamic control mechanism ensures optimal dam operations and minimizes the risk of both flooding and water scarcity.

4. Cost-Efficiency and Sustainability: Implementing an IoT-based flood detection and dam controller system offers long-term cost savings compared to traditional approaches. By leveraging wireless communication and sensor technology, we can reduce the need for costly infrastructure upgrades and manual labour associated with conventional flood monitoring and management. Furthermore, the ability to optimize dam operations leads to more efficient water resource utilization, promoting sustainability in water management practices.

5. Enhancing Resilience and Community Safety: Ultimately, the goal of this initiative is to enhance the resilience of communities living in flood-prone areas and ensure their safety in the face of natural disasters. By harnessing the power of IoT technology, we empower authorities, emergency responders, and residents with timely and accurate information, enabling them to make informed decisions and take proactive measures to mitigate the impacts of floods effectively.

1.3 Statement of problem

The development and implementation of an IoT-based flood detection and dam controller system face several key challenges:

1. Sensor Reliability and Accuracy:
   - IoT sensors deployed for flood detection must be highly reliable and accurate in measuring parameters such as water levels, rainfall intensity, and weather conditions.
   - Sensor calibration, maintenance, and environmental factors (e.g., debris accumulation) can affect the accuracy and reliability of sensor data, leading to erroneous flood predictions and control decisions.
2. Data Transmission and Connectivity:
   - Effective communication between IoT sensors, data processing units, and control centres is essential for real-time monitoring and response.
   - Challenges such as network connectivity issues, signal interference, and data transmission delays can hinder the timely delivery of sensor data, impacting the system's responsiveness and reliability.
3. Data Processing and Analysis:
   - The sheer volume of data generated by IoT sensors requires efficient processing and analysis techniques to extract actionable insights.

- Advanced algorithms for data fusion, anomaly detection, and flood prediction are needed to interpret sensor data accurately and facilitate timely decision-making.

4. System Integration and Interoperability:
    - Integrating IoT-based flood detection and dam control systems with existing infrastructure, such as emergency management systems and water management facilities, presents interoperability challenges.
    - Compatibility issues between different hardware and software components may arise, requiring standardized protocols and interfaces for seamless integration.
5. Security and Privacy Concerns:
    - IoT devices and communication networks are susceptible to cybersecurity threats such as hacking, data breaches, and unauthorized access.
    - Ensuring the security and privacy of sensor data, control commands, and system operations is paramount to prevent malicious attacks and safeguard sensitive information.
6. Regulatory and Environmental Considerations:
    - Compliance with regulatory requirements and environmental regulations, such as water resource management policies and environmental impact assessments, is essential for the sustainable deployment and operation of IoT-based flood management systems.
    - Balancing the need for flood control with environmental conservation and ecosystem preservation requires careful planning and stakeholder engagement.

1.4 Objective of the study

1. Design and Deployment of IoT Sensor Network:
    - Develop a comprehensive plan for deploying IoT sensors in flood-prone areas to monitor water levels, rainfall intensity, weather conditions, and other relevant parameters.
    - Design sensor nodes capable of reliable and accurate data collection, considering factors such as sensor type, placement, power source, and communication protocols.
2. Real-Time Data Acquisition and Processing:
    - Implement data acquisition systems to collect sensor data in real-time and transmit it to a centralized processing unit.
    - Develop algorithms and data processing techniques for analysing sensor data, identifying flood events, and predicting flood patterns with high accuracy and reliability.
3. Automated Dam Control and Water Release:
    - Design and integrate automated control mechanisms into dam infrastructure to regulate water release based on real-time sensor data and predefined control algorithms.
    - Implement safety measures and fail-safe mechanisms to prevent potential hazards and ensure the optimal operation of dam control systems.
4. System Integration and Interoperability:
    - Develop interfaces and protocols for seamless integration of the IoT-based flood detection and dam controller system with existing flood management infrastructure, emergency response systems, and water resource management facilities.
    - Ensure interoperability with different hardware and software components, allowing for efficient data exchange and collaboration among stakeholders.
5. Evaluation and Validation:

- Conduct comprehensive field tests and simulations to evaluate the performance, reliability, and effectiveness of the IoT-based flood detection and dam controller system under various scenarios and environmental conditions.
- Validate the accuracy of flood detection algorithms, the efficiency of automated dam control mechanisms, and the overall resilience of the system in mitigating flood impacts.

6. Stakeholder Engagement and Knowledge Transfer:
    - Engage with relevant stakeholders, including government agencies, local communities, academic institutions, and technology providers, throughout the project lifecycle to gather feedback, address concerns, and ensure the adoption of the proposed solution.
    - Disseminate knowledge and best practices derived from the study through workshops, training programs, and knowledge-sharing platforms to empower stakeholders with the necessary skills and information for effective flood management.

Chapter-2(literature survey)

2.1 literature survey

In recent years, significant strides have been made in leveraging IoT technology for flood detection and dam control systems. Researchers and practitioners have explored a multitude of sensor types and data processing techniques to enhance the accuracy and reliability of flood monitoring. Water level sensors, rain gauges, and weather stations, among other IoT devices, have been strategically deployed in flood-prone areas to collect real-time data on hydrological parameters. These sensors, often integrated into sensor networks, enable continuous monitoring of water levels in rivers, streams, and reservoirs, providing valuable insights into flood dynamics. Moreover, advancements in data processing and analytics have facilitated the development of sophisticated algorithms for flood prediction and risk assessment. Machine learning algorithms, coupled with remote sensing data and hydrological models, offer improved spatial and temporal resolution, aiding in the early identification of flood events and the estimation of their magnitude. Additionally, the integration of IoT technology into dam control systems has revolutionized water management practices, allowing for adaptive control strategies based on sensor data and predictive analytics. Automated dam control mechanisms, guided by algorithms such as fuzzy logic and model predictive control, optimize water release operations to mitigate flood risks while ensuring efficient water resource utilization. However, despite these advancements, challenges such as sensor reliability, data integration, interoperability, and cybersecurity remain significant hurdles in the development and deployment of IoT-based flood detection and dam controller systems. Addressing these challenges and exploring emerging technologies hold the key to realizing the full potential of IoT in revolutionizing flood management strategies.

2.2 Theoretical Background

An IoT-based flood detection and dam controller system draws upon principles from hydrology, sensor technology, data analytics, and control systems engineering. Hydrological concepts provide insight into water flow dynamics and flood behaviour, while sensor technology enables the collection of real-time data on water levels, rainfall, and weather conditions. Data analytics techniques process and analyse sensor data for flood prediction and risk assessment, utilizing machine learning algorithms and statistical models. Control systems engineering principles guide the design of automated dam control systems, optimizing water release operations based on sensor feedback to mitigate flood risks effectively. Integrating these theoretical foundations

enhances the resilience of flood management strategies in vulnerable areas. Ongoing research and innovation in these fields are vital for advancing the capabilities of IoT-based flood detection and dam control technologies.

Chapter -3 (Scope of project)

The IoT-based flood detection and dam controller system project encompasses several key aspects, including:

1. Sensor Deployment and Network Design:
   - Identifying flood-prone areas and determining optimal locations for sensor deployment.
   - Selecting appropriate sensors for measuring water levels, rainfall intensity, weather conditions, and other relevant parameters.
   - Designing a robust sensor network architecture for seamless data collection and transmission.

2. Data Acquisition and Processing:
   - Developing systems for real-time data acquisition from IoT sensors.
   - Implementing data processing algorithms to analyse sensor data and identify flood events.
   - Integrating remote sensing data and hydrological models to enhance flood prediction accuracy.

3. Automated Dam Control Mechanisms:
   - Designing automated control systems for dam operations based on sensor data and predictive analytics.
   - Implementing control algorithms for optimizing water release operations to mitigate flood risks.
   - Ensuring the safety and reliability of automated dam control mechanisms through fail-safe measures.

4. System Integration and Interoperability:
   - Integrating the IoT-based flood detection and dam controller system with existing infrastructure, including emergency management systems and water resource management facilities.
   - Establishing interoperability with different hardware and software components to facilitate seamless data exchange and communication.
   - Ensuring compatibility with standardized protocols and interfaces for efficient system integration.

5. Evaluation and Validation:
   - Conducting field tests and simulations to evaluate the performance and reliability of the system under various environmental conditions.
   - Validating the accuracy of flood detection algorithms and the effectiveness of automated dam control mechanisms.
   - Gathering feedback from stakeholders and end-users to assess system usability, functionality, and overall satisfaction.

6. Stakeholder Engagement and Knowledge Transfer:
- Engaging with government agencies, local communities, and other stakeholders throughout the project lifecycle.
- Providing training and support to stakeholders on system operation, maintenance, and emergency response procedures.
- Disseminating project findings, best practices, and lessons learned through workshops, conferences, and knowledge-sharing platforms.

By addressing these components within the scope of the project, the IoT-based flood detection and dam controller system aims to enhance flood management strategies, improve community resilience, and mitigate the impact of floods on lives and infrastructure.

Chapter 4 (System design)

4.1 Block diagram

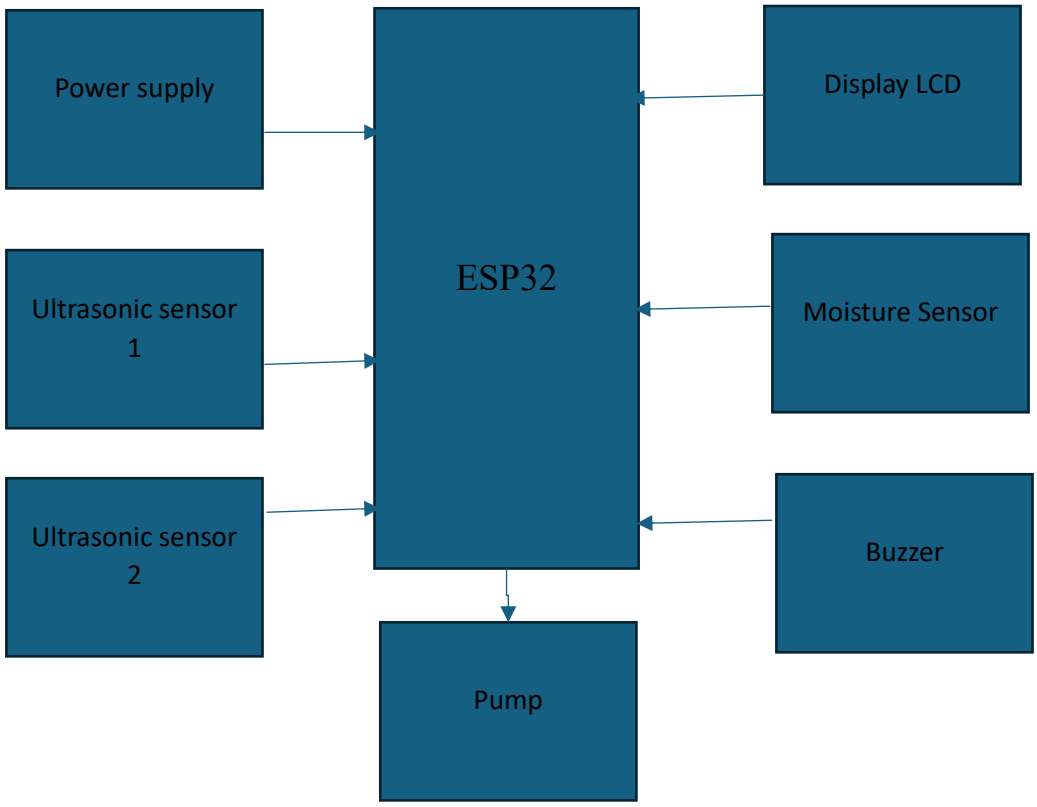


Fig 1. Block Diagram of system

IoT Early Flood Detection & Avoidance System" is an intelligent system which keeps close watch over various natural factors to predict a flood, so we can embrace ourselves for caution, to minimise the damage caused by the flood. Natural disasters like a flood can be devastating leading to property damage and loss of lives. To eliminate or lessen the impacts of the flood, the system uses various natural factors to detect flood. The system has a Wi-Fi connectivity, thus its collected data can be accessed from anywhere quite easily using IoT.

To detect a flood the system observes various natural factors, which includes humidity, temperature, water level and flow level. To collect data of mentioned natural factors the system consist of different sensors which collects data for individual parameters. For detecting changes in humidity and temperature the system has a Digital Temperature Humidity Sensor. It is an advanced sensor module with consists of resistive humidity and temperature detection components. The water level is always under observation by a float sensor, which work by opening and closing circuits (dry contacts) as water levels rise and fall. It normally rest in the

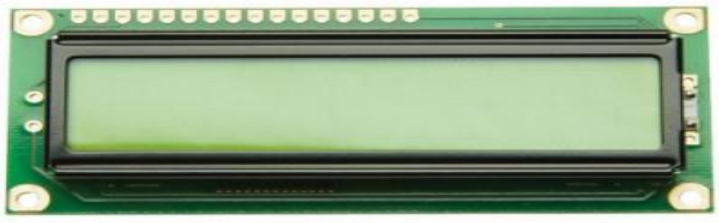
closed position, meaning the circuit is incomplete and no electricity is passing through the wires yet. Once the water level drops below a predetermined point, the circuit completes itself and sends electricity through the completed circuit to trigger an alarm. The flow sensor on the system keeps eye on the flow of water. The water flow sensor consists of a plastic valve body, a water rotor, and a hall-effect sensor. When water flows through the rotor, rotor rolls. Its speed changes with different rate of flow.

The system also consist of a HC-SR04 Ultrasonic Range Finder Distance Sensor. The Ultrasonic sensor works on the principle of SONAR and is designed to measure the distance using ultrasonic wave to determine the distance of an object from the sensor. All the sensors are connected to Arduino UNO, which processes and saves data. The system has Wi-Fi feature, which is useful to access the system and its data over IoT.

4.2 Description

1.LCD 16*2

The 16x2 LCD (Liquid Crystal Display) module is a widely used component in various electronic projects for displaying textual information. With its compact size and simplicity, it provides a user-friendly interface for presenting data. Hardware setup involves connecting the LCD to the Arduino board, ensuring proper wiring of power, ground, and data pins. Utilizing the Liquid Crystal library in the Arduino IDE simplifies the programming process. Initialization of the LCD in the setup () function involves specifying the dimensions of the display, typically 16 columns by 2 rows. From there, text can be printed to the display using the lcd.print () function, with lcd.setCursor () allowing precise positioning of the cursor for displaying text at specific locations. Additional features such as scrolling text, adjusting contrast, or displaying custom characters can be implemented by utilizing appropriate functions and commands provided by the Liquid Crystal



library. With its ease of use and versatility, the 16x2 LCD module serves as an indispensable tool for conveying information in Arduino-based projects.

Fig 2. LCD 16*2

Controller

2. ESP32

The ESP32, developed by Espressif Systems, is a versatile microcontroller featuring a dual-core processor running at up to 240 MHz it offers built-in Wi-Fi and Bluetooth connectivity, making it ideal for IoT applications. With a rich set of peripheral interfaces including UART, SPI, I2C, and ADC, interfacing with external devices is seamless. Despite its high performance, the ESP32 boasts low power consumption, suitable for battery-powered applications. Supported by popular development frameworks like Arduino IDE and Espressif IDF, the ESP32 provides a flexible platform for developing a wide range of IoT projects.



Fig 3. ESP32

3.I2C

I2C, or Inter-Integrated Circuit, is a widely used serial communication protocol for connecting multiple devices in a master-slave configuration. It uses two wires, SDA (data line) and SCL (clock line), to transmit data between devices. I2C supports multiple devices on the same bus, each with a unique address, allowing for efficient communication and control. It is commonly used in embedded systems, sensors, and other electronic applications due to its simplicity, versatility, and low hardware requirements.

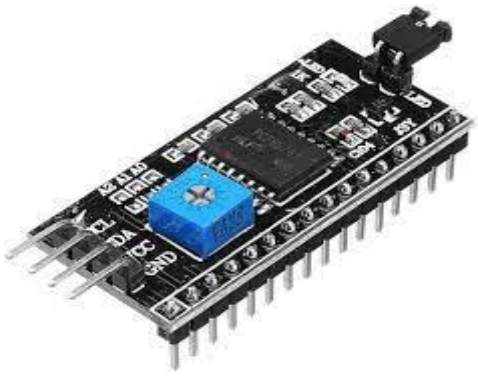


Fig 4. I2C

4.Two Channel Relay Module

A 2-channel relay module is an electronic device used to control high-power circuits or devices with low-power signals. It typically consists of two relays, each capable of independently switching a separate circuit on or off. By applying a digital signal to the relay, it toggles its internal contacts, allowing or interrupting the flow of current through the connected circuit. This module provides electrical isolation between the control signal and the switched circuit, making it suitable for interfacing with sensitive electronics. With its specified voltage and current ratings, the relay module can handle various loads, including lights, motors, solenoids, and valves. It's commonly used in home automation, robotics, and industrial control systems, offering a convenient solution for controlling multiple high-power devices using microcontrollers like Arduino or Raspberry Pi.



Fig 5. 2 Channel Relay Module

5.Jumper wire

A jumper wire is an electric wire that connects remote electric circuits used for printed circuit boards. By attaching a jumper wire on the circuit, it can be short-circuited and short-cut (jump) to the electric circuit.



Fig 6. Jumper wires

6.PCB

A **printed circuit board** is a medium used in electrical and electronic engineering to connect electronic components to one another in a controlled manner. It takes the form of a laminated sandwich structure of conductive and insulating layers: each of the conductive layers is designed with an artwork pattern of traces, planes and other features (similar to wires on a flat surface) etched from one or more sheet layers of copper laminated onto and/or between sheet layers of a non-conductive substrate.[1] Electrical components may be fixed to conductive pads on the outer layers in the shape designed to accept the component's terminals, generally by means of soldering, to both electrically connect and mechanically fasten them to it. Another manufacturing process adds vias: plated-through holes that allow interconnections between layers.



Fig 7. PCB

7.Ultrasonic Sensor

An ultrasonic sensor is a device that uses ultrasonic sound waves to measure distance to or detect the presence of objects. It emits high-frequency sound waves beyond the range of human hearing and measures the time it takes for the sound waves to bounce back after hitting an object. This information is then used to calculate the distance to the object. Ultrasonic sensors are commonly used in robotics, automotive parking assistance systems, industrial automation, and various other applications where accurate distance measurement or object detection is required.



Fig 8. Ultrasonic Sensor

8. LCD

LCD (Liquid Crystal Display) is a type of flat panel display which uses liquid crystals in its primary form of operation. LEDs have a large and varying set of use cases for consumers and businesses, as they can be commonly found in smartphones, televisions, computer monitors and instrument panels.



Fig 9. LCD

9.Moisture Sensor

A moisture sensor is a device used to detect the presence or absence of moisture in a substance or environment. It typically works by measuring the conductivity, capacitance, or resistance of a material, which changes with variations in moisture content. These sensors are commonly used in agriculture for soil moisture monitoring, in industrial processes to control moisture levels in materials, and in consumer products such as humidity sensors in air conditioners and dehumidifiers.

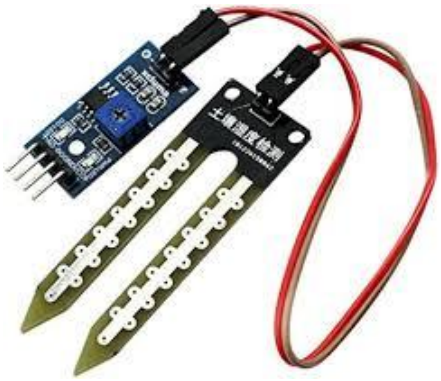


Fig 10. Moisture Sensor

10.Buzzer

A buzzer is an electrical device that produces a buzzing or beeping sound when activated. It typically consists of a vibrating component, such as a metal diaphragm or piezoelectric element, and an electrical circuit that causes the vibration when powered. Buzzer devices are commonly used in alarms, timers, electronic games, and various other applications to indicate a specific event or signal.



Fig 11.Buzzer

11.DC Battery

A DC battery, short for Direct Current battery, is a device that stores electrical energy in the form of direct current. It consists of one or more electrochemical cells, which convert chemical energy into electrical energy through chemical reactions. DC batteries are commonly used in various applications such as automotive vehicles, portable electronic devices, backup power systems, and renewable energy storage systems. They provide a reliable source of power for devices and systems that require a steady supply of electricity.



Fig 12.DC Battery

12.DC Pump

A DC pump is a type of pump that operates using direct current (DC) electrical power. These pumps are commonly used in various applications where a continuous flow of liquid or gas is required, such as in aquariums, water circulation systems, solar water heating systems, and fuel transfer. DC pumps are often favoured for their efficiency, reliability, and ability to be powered by batteries or solar panels, making them suitable for remote or off-grid locations. They come in various sizes and designs to accommodate different flow rates and pressure requirements.



Fig 13. DC Pump

Chapter 5 (System implementation)

5.1. Circuit Diagram:

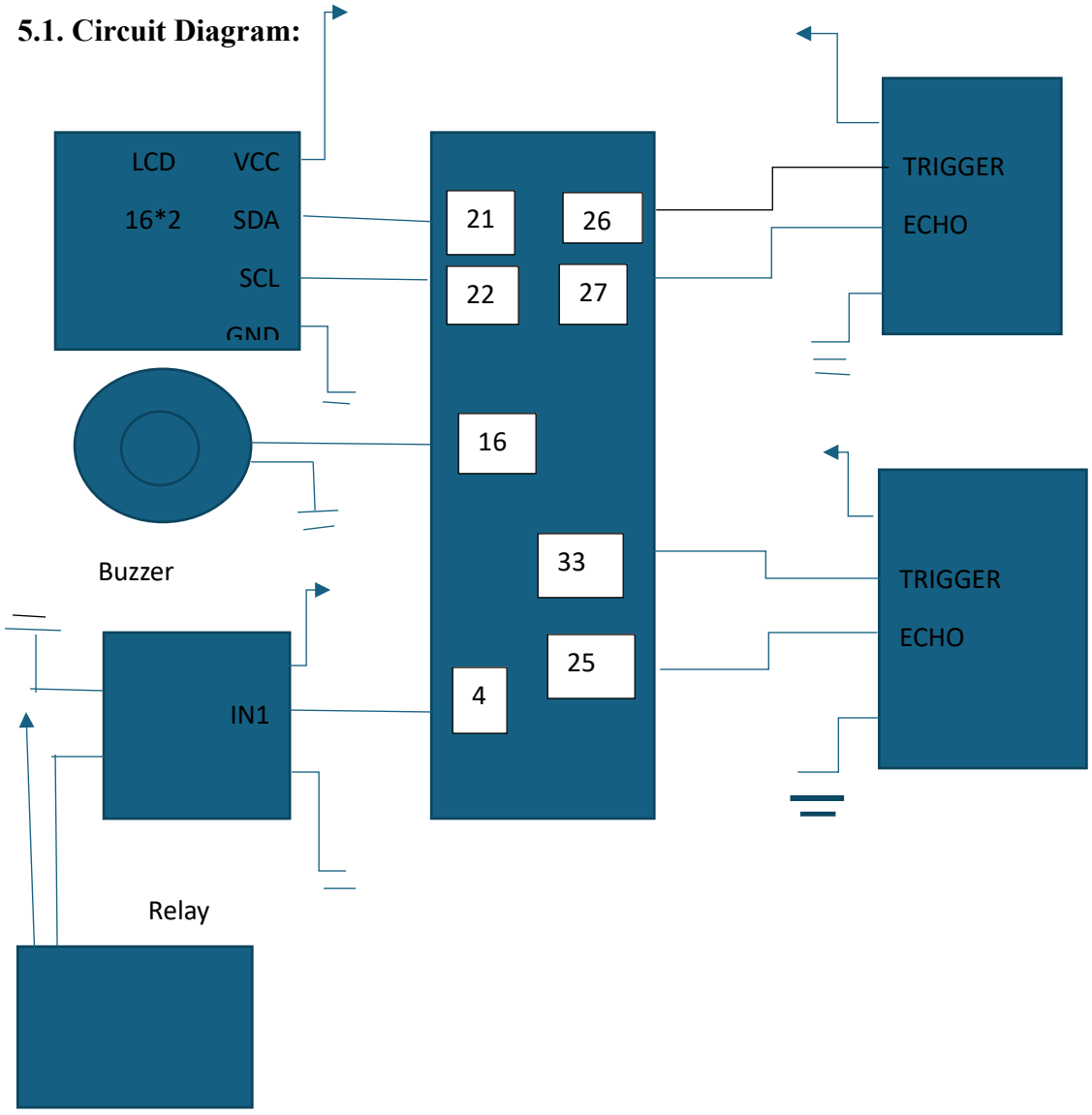


Fig 14. Circuit Diagram

Working of each block:

This consists ESP 32 MC, Ultrasonic Sensor HC-SR04, 2. Relay Module, Moisture Sensor, DC Battery or Pump, 16x2 LCD Display Following the function of each component given below

1. ESP-32:

It is used for to connect the with Wi-Fi module to the device for dual Core processing.

2. Ultrasonic Sensor:

It is used to measure distance of Sound waves of water level to sense the water.

3. 2 Relay module:

It is used to control the high voltage with high current, of Arduino & control the Power of arduino.

4. Moisture sensor:

It is used to detects the precipitation Such as rain or snow by measuring presence of water on its Surface

5. DC Battery:

It is convert Used Supply to the Device & the convert chemical energy into electrical energy through chemical reaction.

6. DC Pump:

It operates by converting direct current electrical energy into mechanical energy this typically. Achieved through the use of motor that spins abs impeller. Flow of liquid creating a flow of liquid

7.16*2 LCD Display:

It is used to display character to the microcontroller sends commands data to the display's pixels the controller interprets the data & generate the & appropriate Signals to display characters at specified position of the screen

5.2. Flowchart

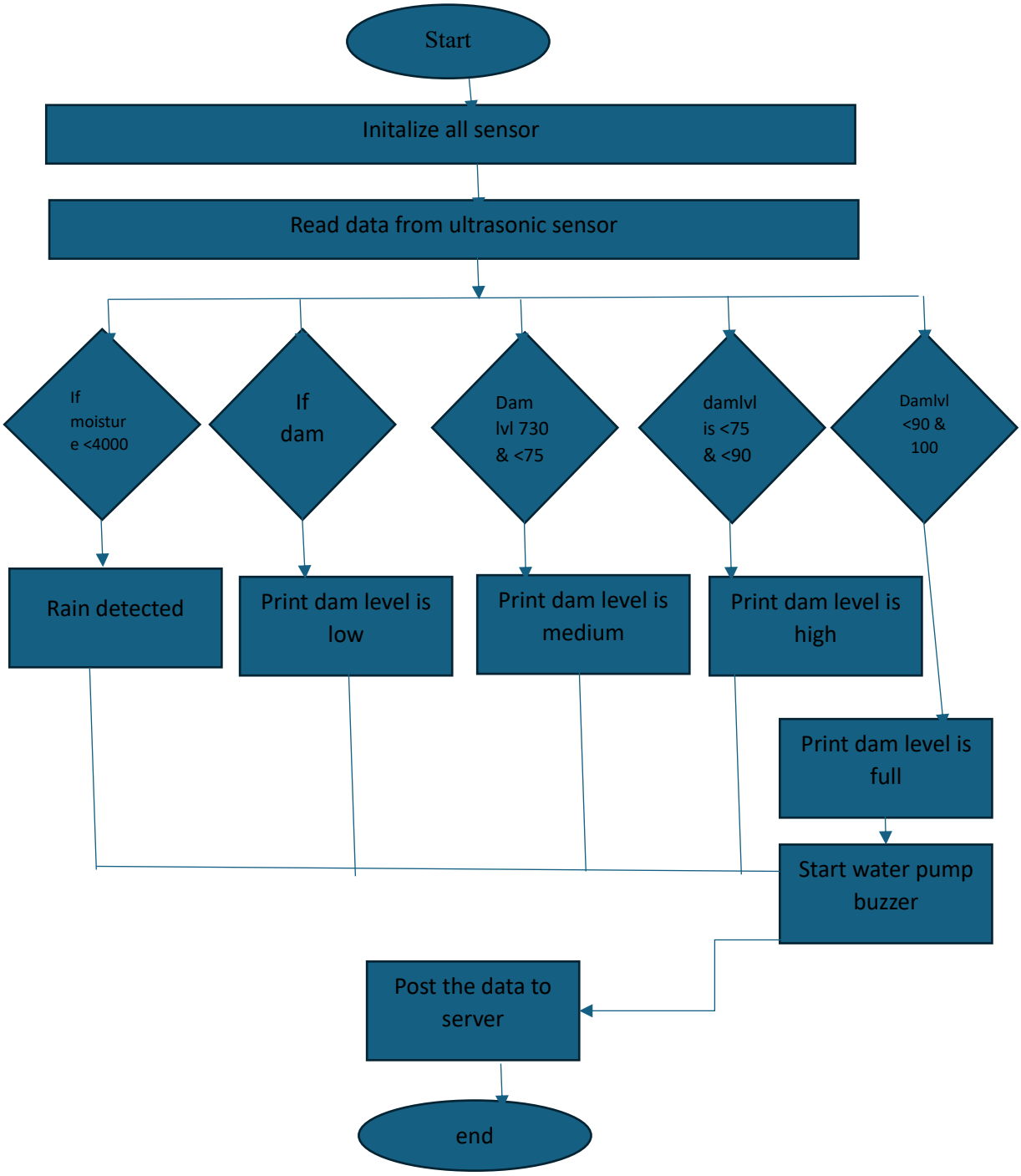


Fig 15. Flowchart

5.3. APP

We Developed an App for alerting the common people about the Dam level in four stages as follows low, medium, high, and full.
When the dam level is low i.e., between 0 to 30
When the dam level is medium i.e., between 30 to 75
When the dam level is high i.e., between 75 to 90
When the dam level is full ice, between 90 to 100

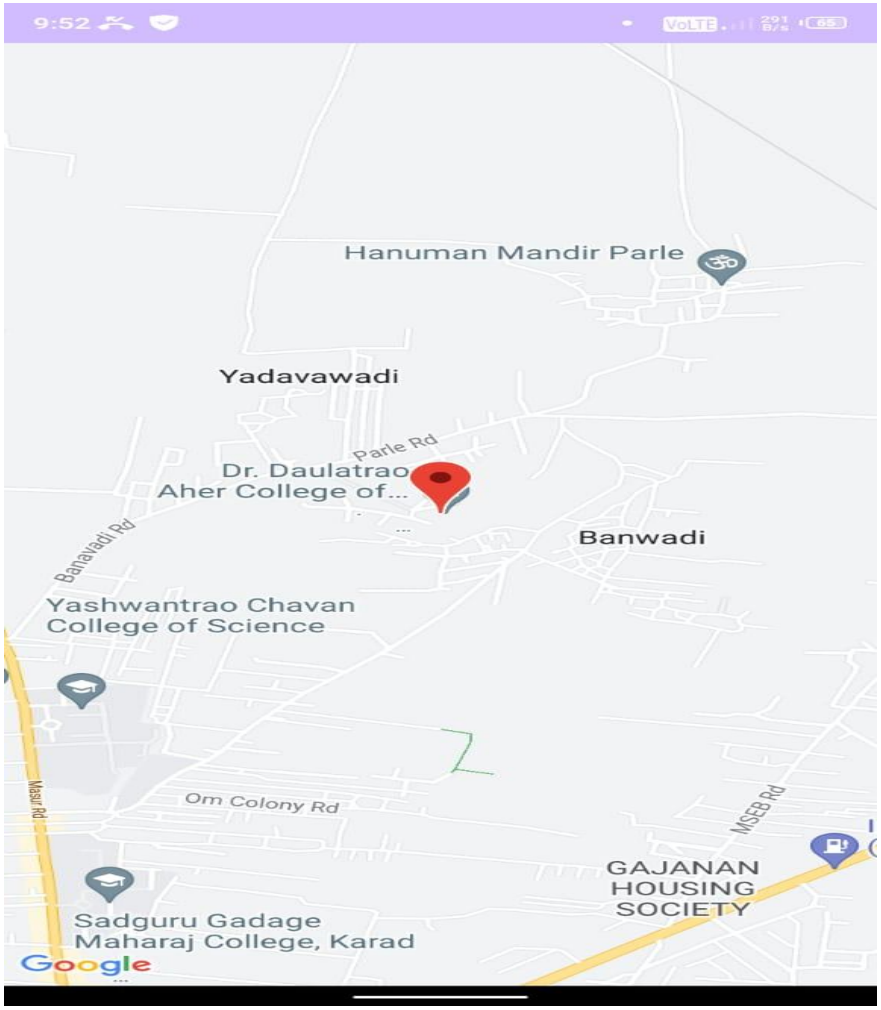


Fig 16. Structure of APP



Fig 17 .Alert notification from APP.

5.2 Details/ Procedure of software tools use:

a. Arduino IDE software



Fig 18. Arduino IDE software

Arduino IDE software Arduino IDE Software information:

Arduino IDE (Integrated Development Environment) is a software program used to write and upload code to Arduino boards. It is an open-source software that supports both Windows, macOS, and Linux operating systems.

The IDE provides a user-friendly interface to write, compile, and upload code to Arduino boards. It includes a code editor with syntax highlighting, a serial monitor for debugging, and a library manager to easily add pre-built libraries to your project.

Arduino IDE uses the C++ programming language and a simplified version of it called Wiring. The code is compiled and uploaded to the Arduino board through a USB connection. Overall, Arduino IDE is a popular and easy-to-use software for beginners and professionals to create projects with Arduino boards.

Arduino IDE Software working:

Arduino IDE works by providing a user-friendly interface for writing and uploading code to Arduino boards. Here's a simplified overview of how it works: Writing code: Users can write code in the Arduino IDE's code editor, which includes syntax highlighting and auto-completion features to help with coding.

The code is written using the C++ programming language or a simplified version called Wiring. Compiling code: Once the code is written, the user can compile it using the "Verify" button in the IDE. The IDE then checks the code for errors and generates a binary file.

24 Uploading code: The user can upload the compiled binary file to the Arduino board using the "Upload" button in the IDE. The board must be connected to the computer via USB for this step to work.

Running code: Once the code is uploaded to the board, it will start running automatically. The user can monitor the board's output using the IDE's serial monitor, which displays data sent from the board over the serial port.

Overall, Arduino IDE provides a simplified and streamlined way for users to write and upload code to Arduino boards, making it an ideal choice for beginners and professionals alike.

b. MIT App inventor:



Fig 19. MIT APP INVENTOR

MIT App Inventor MIT App Inventor is a free, cloud-based platform that allows users to create mobile apps for Android devices using a visual programming language. It provides a simple and intuitive drag and-drop interface for designing the user interface and programming the app's behaviour.

5.3 programming

A. Code

```
#ifdef ESP32
  #include <WiFi.h>
  #include <HTTPClient.h>

#else
  #include <ESP8266WiFi.h>
  #include <ESP8266HTTPClient.h>
  #include <WiFiClient.h>
#endif

#include <Wire.h>
//Access point credentials

#include<LiquidCrystal_I2C.h>
LiquidCrystal_I2C lcd(0x27, 16, 2);

const char* ssid = "Mahek";
const char* pwd = "mahek123";
const char* host = "http://tsm.ecssofttech.com";
String get_host = "http://tsm.ecssofttech.com/";

WiFiServer server (80); // open port 80 for server connection


int trig1 = 26;
int echo1 = 27;
int trig2 = 33;
int echo2 = 25;
int buz = 16;
int rain = 34;
int pump = 4;
int state1 = 0;
long duration1, cm1,duration2, cm2;

void setup() {
  //Serial.begin(9600);
  lcd.init();
  lcd.backlight ();
  lcd.setCursor (0, 0);
  lcd.print("IoT Based Flood");
```

```
  lcd.setCursor(0, 1);
  lcd.print("Detect & DamCtrl");
  delay(5000);
  lcd.clear();

  pinMode(trig1, OUTPUT);
  pinMode(echo1, INPUT);
  pinMode(trig2, OUTPUT);
  pinMode(echo2, INPUT);
  pinMode(buz, OUTPUT);
  pinMode(pump, OUTPUT);
  digitalWrite(pump, HIGH);
  WiFi.begin(ssid, pwd);
  server.begin();


  Serial.begin(9600);
}

void loop() {
sensor1();
sensor2();
}

void sensor1()
{

  digitalWrite(trig1, LOW);
delayMicroseconds(5);
digitalWrite(trig1, HIGH);
delayMicroseconds(5);
digitalWrite(trig1, LOW);
duration1 = pulseIn(echo1, HIGH);
cm1 = (duration1/2)/20.9;
Serial.print("distance1 ");
Serial.println(cm1);
if(cm1 > 20 && cm1 < 400)
{
  digitalWrite(pump, HIGH);
  lcd.setCursor(0,0);
  lcd.print("Dam Level is");
  lcd.setCursor(0, 1);
  lcd.print("Less Than 50%");
```

```
 delay(2000);
 lcd.clear();
 state1 = 0;

 }
 else if(cm1 <= 20 && cm1 > 5)
 {
    state1 = 0;
    float per = cm2 *(100/35);
    lcd.setCursor(0,0);
    lcd.print("Dam Level is");
    lcd.setCursor(0, 1);
    lcd.print(100-per);
    lcd.setCursor(6, 1);
    lcd.print("% Full");
    digitalWrite(pump, HIGH);
    delay(2000);
    lcd.clear();

 }

 else if(cm1 <= 5 && cm1 >= 0)
 {
    state1 = 1;
    digitalWrite(pump, LOW);
    lcd.setCursor(0,0);
    lcd.print("Dam Level is");
    lcd.setCursor(0, 1);
    lcd.print("almost full");
    get_device_status();
    delay(2000);
    lcd.clear();
 }

}
 void sensor2()
{

digitalWrite(trig2, LOW);
delayMicroseconds(5);
digitalWrite(trig2, HIGH);
delayMicroseconds(5);
digitalWrite(trig2, LOW);
```

```
duration2 = pulseIn(echo2, HIGH);
cm2 = (duration2/2)/20.9;
Serial.print("distance2 ");
Serial.println(cm2);
delay(1000);
if(cm2 > 20 && cm2 < 400)
{
 digitalWrite(pump, HIGH);
 digitalWrite(buz, LOW);
 lcd.setCursor(0,0);
 lcd.print("Dam Level is");
 lcd.setCursor(0, 1);
 lcd.print("Less Than 50%");
 delay(2000);
 lcd.clear();
 state1 = 0;

 }
 else if(cm2 <= 20 && cm2 > 5)
 {
    digitalWrite(pump, HIGH);
    float per = cm2 *(100/35);
    lcd.setCursor(0,0);
    lcd.print("Dam Level is");
    lcd.setCursor(0, 1);
    lcd.print(100-per);
    lcd.setCursor(6, 1);
    lcd.print("% Full");
    delay(2000);
    digitalWrite(buz, LOW);
    lcd.clear();
    state1 = 0;
 }

 else if(cm2 <= 5 && cm2 >= 0)
 {
    digitalWrite(pump, LOW);
    digitalWrite(buz, HIGH);
    lcd.setCursor(0,0);
    lcd.print("Dam Level is");
    lcd.setCursor(0, 1);
    lcd.print("almost full");
    delay(2000);
```

```
    lcd.clear();
    state1 = 1;
    get_device_status();
  }
}

void get_device_status()
  {
```

Chapter-6 (Result and Applications)

6.1 Result

The project is an output of 1 year of research and implementation. The circuits when implemented separately works as per the desired output however during integrating all, output fluctuates and shows different response every time. This could be a problem of loose connections of the wires or internal wiring of the bread board used. This project lists down the results realized from the practical work and examines whether ideas / solution approaches recommended in research are met by the practical implementation.

we have conducted the following results were obtained:

- Early Warning and Rapid Response.
- Improved Monitoring and Management.
- Enhanced Public Safety.

6.2 Application

An IoT-based flood detection and dam controller system utilizes sensors to monitor water levels and rainfall intensity in flood-prone areas. Real-time data is analyzed to detect potential flood risks, triggering alerts to authorities. Dam controllers adjust gates or release valves based on incoming data to regulate water flow and mitigate downstream flooding. Historical data is stored for analysis, aiding in forecasting future events. Remote access allows stakeholders to monitor and control operations from anywhere. Integration with existing infrastructure enhances overall effectiveness in managing flood risks, protecting lives, and safeguarding critical infrastructure.

Chapter -7 (Conclusion and Future scope)

7.1 Conclusion

In this project I have implemented IOT Based flood detection & dam controller system. The hardware equipment is tested and result is obtained. This project is cost effective. Implementation of this project in present day will effectively solve the flood problem which is a severe problem in many areas of all over the world.

An IoT-based flood detection and dam controller system offers real-time monitoring, predictive analytics, and remote control capabilities to effectively manage flood risks and optimize dam operations, enhancing safety and resilience in flood-prone areas. In practice presently in India we are following time-based control on traffic signals and we are experiencing a heavy traffic jam all over which in turn consumes lot of time and fuel. We hope this method will be adopted as soon as possible so that the limitations we are experiencing with present method can be overcome .

7.2 Advantage

1. Early Warning: IoT-based flood detection provides early warnings, enabling timely evacuation and mitigation measures.

2. Real-Time Monitoring: Continuous monitoring of water levels and weather conditions allows for immediate response to changing flood risks.

3. Remote Management: Dam controllers can be operated remotely, enhancing operational efficiency and safety for dam personnel.

4. Data-Driven Decision Making: Access to real-time and historical data supports informed decision-making for flood management strategies.

5. Cost-Efficiency: IoT technology reduces the need for manual monitoring and maintenance, leading to cost savings in flood management efforts.

7.3 Limitation of the System

IoT-based flood detection and dam controller systems may face limitations such as reliability concerns related to internet and power dependencies, potential inaccuracies in data collection, cybersecurity vulnerabilities, high implementation costs, integration challenges, and coverage gaps in monitoring areas.

7.4 Future scope

In envisioning the future of IoT-based flood detection and dam control systems, a landscape of dynamic advancements emerges, promising transformative impacts on flood management and disaster resilience. Predictive analytics, bolstered by machine learning and AI algorithms, are poised to refine flood event forecasting, empowering preemptive interventions well ahead of impending disasters. Moreover, integrating these systems with climate models and weather forecasts augments our understanding of long-term weather patterns, thereby fortifying decision-making processes and risk mitigation strategies. As smart infrastructure becomes more prevalent, the integration of IoT sensors directly into buildings, roads, and other critical assets enables real-time monitoring and automated responses to flood risks. Embracing multi-scale monitoring, from individual structures to entire watersheds, promises a holistic view of flood vulnerabilities, facilitating tailored interventions at various levels. Collaboration between governments, research institutions, and private sectors fosters data sharing and resource pooling, enriching flood management efforts. Furthermore, community engagement initiatives empower local populations through citizen science and participatory sensing programs, fostering proactive measures to safeguard communities against flooding. Integrating IoT-based systems into resilient infrastructure designs strengthens climate adaptation strategies, enhancing societal resilience to extreme weather events. Robust policy frameworks ensure ethical deployment and operation of IoT technologies, safeguarding privacy and security concerns. International collaboration drives global solutions to flooding challenges, promoting knowledge exchange and technology transfer. Recognizing the broader ecosystem services facilitated by these systems underscores their role in preserving biodiversity, protecting natural habitats, and promoting sustainable water management practices. In sum, the future of IoT-based flood detection and dam control systems holds immense promise for fostering safer, more resilient communities amidst the escalating threats of climate change and natural disasters.

Chapter-8 (Reference and Bibliography)

8.1. Reference

https://ieeexplore.ieee.org/abstract/document/5713489

Deep Learning-based **System** for Change **Detection** On-Board Earth Observation Small Satellites

Impact of alternate wetting and drying irrigation on arsenic uptake and speciation in **flooded** rice **systems**

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2. Gupta, R., Sharma, S., & Patel, D. (2019). "Wireless Sensor Networks for Flood Monitoring and Dam Control: A Survey." IEEE Transactions on Sustainable Computing, 4(2), 154-168.
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Chapter 9 (Appendix)

9.1 Bill of material

Sr no	Component name	Quantity	Price
1	ESP32	1	550rs
2	Jumper wires	As per requirement	100rs
3	I2C LCD	1	250rs
4	Two relay module	1	150rs
5	PCB	1	20rs
6	Buzzer	1	25rs
7	Male burg strip	1	20rs
8	Female burg strip	1	20rs
9	Ultrasonic Sensor	2	80rs
10	Moisture Sensor	1	100rs
11	DC Pump	1	90rs

12	DC Battery	1	100rs

9.2 Datasheets

1. ESP32

A. Pin Layout:

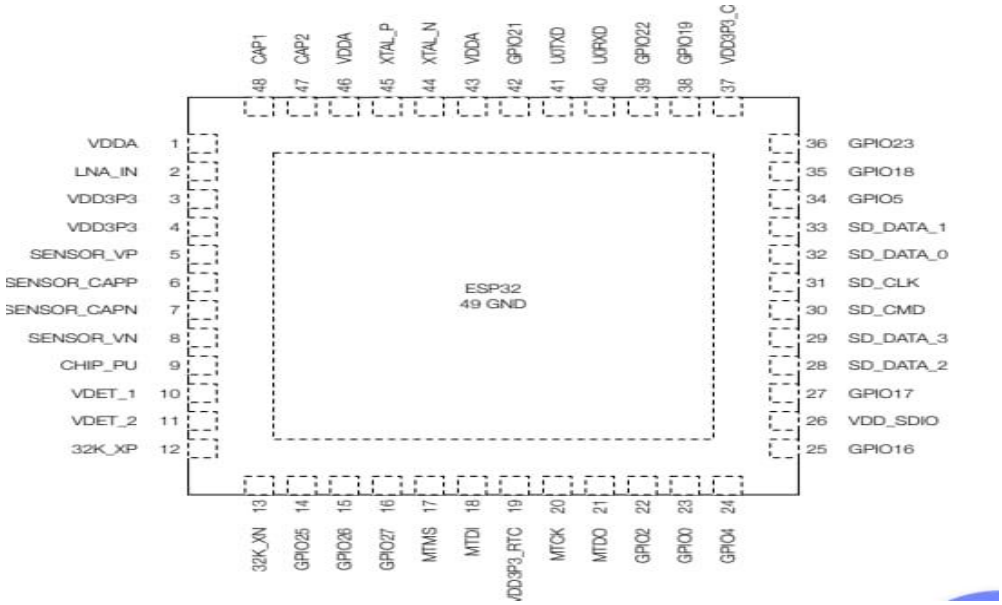


Fig 20. Pin Layout

ESP32 Peripherals and I/O

Although the ESP32 has 48 GPIO pins in total, only 25 of them are broken out to the pin headers on both sides of the development board. These pins can be assigned a variety of peripheral duties, including:

15 ADC channels	15 channels of 12-bit SAR ADC with selectable ranges of 0-1V, 0-1.4V, 0-2V, or 0-4V

2 UART interfaces	2 UART interfaces with flow control and IrDA support
25 PWM outputs	25 PWM pins to control things like motor speed or LED brightness
2 DAC channels	Two 8-bit DACs to generate true analogy voltages
SPI, I2C and I2S interface	Three SPI and one I2C interfaces for connecting various sensors and peripherals, as well as two I2S interfaces for adding sound to your project
9 Touch Pads	9 GPIOs with capacitive touch sensing

B. Wi-Fi

ESP32 implements a TCP/IP and full 802.11 b/g/n Wi-Fi MAC protocol. It supports the Basic Service Set (BSS) STA and Sofa operations under the Distributed Control Function (DCF). Power management is handled with minimal host interaction to minimize the active-duty period.

Wi-Fi Radio and Baseband

The ESP32 Wi-Fi Radio and Baseband support the following features:

802.11b/g/n

802.11n MCS0-7 in both 20 MHz and 40 MHz bandwidth

802.11n MCS32 (RX)

802.11n 0.4 με quard-interval

up to 150 Mbps of data rate

Receiving STBC 2x1

Up to 20.5 dBm of transmitting power

Adjustable transmitting power

Antenna diversity

ESP32 supports antenna diversity with an external RF switch. One or more GPIOs control the RF switch and selects the best antenna to minimize the effects of channel fading.

Wi-Fi MAC

The ESP32 Wi-Fi MAC applies low-level protocol functions automatically. They are as follows:

4x virtual Wi-Fi interfaces

Simultaneous Infrastructure BSS Station mode/SoftAP mode/Promiscuous mode

RTS protection, CTS protection, Immediate Block ACK

Defragmentation

TX/RX A-MPDU, RX A-MSDU

TXOP

WMMCCMP (CBC-MAC, counter mode), TKIP (MIC, RC4), WAPI (SMS4), WEP (RC4) and CRC

C. Serial communication

UART

Universal Asynchronous Receive Transmit (UART) or Serial communication is one of the most simple communication protocols between two devices. It transfers data between devices by connecting two wires between the devices, one is the transmission line while the other is the receiving line. The data transfers bit by bit digitally in form of bits from one device to another. The main advantage of this communication protocol is that it's not necessary for both the devices to have the same operating frequency. For example, two microcontrollers operating at different clock frequencies can communicate with each other easily via serial communication. However, a predefined bit rate that is referred to as baud rate usually set in the flash memory of both microcontrollers for the instruction to be understood by both the devices.

Transmitting and receiving serial data

The transmitting UART takes bytes of data and transmits the bits in a sequential form. The second transmitter which is the receiver reassembles the bits into a complete byte. Serial transmission of data through a single wire is actually more cost-effective than parallel transmission through multiple wires.

Communication between two UART devices may be simplex, full-duplex or half-duplex. Simplex communication is a one-direction type of communication where the signal moves from one UART to another. It doesn't have provision for the receiving UART to send back signals. A full-duplex is when both devices can transmit and receive communications at the same time. A half-duplex is when devices take turns to transmit and receive.

Structure of a UART Protocol

A UART contains a clock generator. This allows the sampling in a bit period.

It also contains input and output shift registers.

Transmitting and receiving control.

A read or write control logic.

Other optional components of a UART are: transmit or receive buffers, FIFO buffer memory, and a DMA controller.